



Stata 19.0
SE--Standard Edition

Statistics and Data Science

Copyright 1985-2025 StataCorp LLC
StataCorp
4905 Lakeway Drive
College Station, Texas 77845 USA
800-782-8272 <https://www.stata.com>
979-696-4600 service@stata.com

Stata license: Single-user perpetual
Serial number: 401906345068
Licensed to: SCU
SCU

Notes:

1. Unicode is supported; see [help unicode advice](#).
2. Maximum number of variables is set to 5,000 but can be increased; see [help set maxvar](#).

```
1 . doedit "C:\Users\b7869\Desktop\scu 114-2\多變量分析\期末\bodybattery_analysis.do"
2 . do "C:\Users\b7869\AppData\Local\Temp\STD3ba0_000000.tmp"
3 . clear all
4 .
5 . import delimited "validation_clean_dataset_final.csv", clear
   (encoding automatically selected: UTF-8)
   (18 vars, 222 obs)
6 .
7 . * 查看資料
8 . describe
```

Contains data
Observations: **222**
Variables: **18**

Variable name	Storage type	Display format	Value label	Variable label
id	str36	%36s		
date	str9	%9s		
steps	float	%9.0g		
active_kilocalories	float	%9.0g		
active_time	float	%9.0g		
resting_heart_rate	float	%9.0g		
average_heart_rate	float	%9.0g		
sleep_hours	float	%9.0g		
deep_sleep_ratio	float	%9.0g		
rem_ratio	float	%9.0g		
awake_ratio	float	%9.0g		
sleep_efficiency	float	%9.0g		
sedentary_ratio	float	%9.0g		
active_ratio	float	%9.0g		
highly_active_ratio	float	%9.0g		
body_battery_level	str8	%9s		
total_sport_hours	float	%9.0g		
stress_hours	float	%9.0g		

Sorted by:

Note: Dataset has changed since last saved.

9 . tab body_battery_status

body_battery_status	Freq.	Percent	Cum.
GOOD	80	36.04	36.04
HIGH	43	19.37	55.41
LOW	70	31.53	86.94
VERY_LOW	29	13.06	100.00
Total	222	100.00	

10 .

11 . * 建立 HIGH vs LOW 群組

12 . gen bb_group=.

(222 missing values generated)

13 . replace bb_group=1 if body_battery_status=="HIGH"

(43 real changes made)

14 . replace bb_group=0 if body_battery_status=="LOW"

(70 real changes made)

15 .

16 . * t-test

17 . ttest sleep_hours, by(bb_group)

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	70	6.626151	.2113051	1.767905	6.204609	7.047693
1	43	6.314005	.2546107	1.669594	5.80018	6.82783
Combined	113	6.50737	.1627753	1.730326	6.184851	6.829888
diff		.3121456	.3354626		-.3525959	.9768872

diff = mean(0) - mean(1) t = 0.9305
H0: diff = 0 Degrees of freedom = 111

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.8229 Pr(|T| > |t|) = 0.3541 Pr(T > t) = 0.1771

18 . ttest deep_sleep_ratio, by(bb_group)

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	70	.2189801	.0101738	.0851199	.198684	.2392763
1	43	.2181064	.0151651	.0994444	.1875019	.2487108
Combined	113	.2186476	.0085042	.0904008	.2017977	.2354976
diff		.0008737	.0175943		-.0339905	.0357379

diff = mean(0) - mean(1) t = 0.0497
H0: diff = 0 Degrees of freedom = 111

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
 Pr(T < t) = **0.5198** Pr(|T| > |t|) = **0.9605** Pr(T > t) = **0.4802**

19 . ttest steps, by(bb_group)

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	70	2779.34	201.7383	1687.863	2376.884	3181.797
1	43	2789.381	276.1948	1811.13	2231.998	3346.765
Combined	113	2783.161	162.5355	1727.777	2461.118	3105.205
diff		-10.04105	336.2709		-676.3843	656.3022

diff = mean(0) - mean(1) t = **-0.0299**
 H0: diff = 0 Degrees of freedom = **111**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
 Pr(T < t) = **0.4881** Pr(|T| > |t|) = **0.9762** Pr(T > t) = **0.5119**

20 . ttest average_heart_rate, by(bb_group)

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	70	75.32722	1.010889	8.457702	73.31055	77.34389
1	43	70.34557	1.257418	8.245444	67.80799	72.88314
Combined	113	73.43155	.8172193	8.68716	71.81233	75.05076
diff		4.981649	1.623295		1.764982	8.198317

diff = mean(0) - mean(1) t = **3.0689**
 H0: diff = 0 Degrees of freedom = **111**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
 Pr(T < t) = **0.9986** Pr(|T| > |t|) = **0.0027** Pr(T > t) = **0.0014**

21 .

22 . * ANOVA

23 . encode body_battery_status, gen(bb_status)

24 .

25 . oneway sleep_hours bb_status

Source	Analysis of variance			F	Prob > F
	SS	df	MS		
Between groups	23.9856464	3	7.99521546	2.57	0.0551
Within groups	677.982559	218	3.11001174		
Total	701.968206	221	3.17632672		

Bartlett's equal-variances test: chi2(3) = **5.5732** Prob>chi2 = **0.134**

26 . oneway deep_sleep_ratio bb_status

Source	Analysis of variance			F	Prob > F
	SS	df	MS		
Between groups	.041518953	3	.013839651	1.67	0.1739
Within groups	1.80425938	218	.008276419		
Total	1.84577833	221	.008351938		

Bartlett's equal-variances test: chi2(3) = 2.3412 Prob>chi2 = 0.505

27 . oneway steps bb_status

Source	Analysis of variance			F	Prob > F
	SS	df	MS		
Between groups	2636633.64	3	878877.878	0.27	0.8446
Within groups	701074397	218	3215937.6		
Total	703711030	221	3184212.81		

Bartlett's equal-variances test: chi2(3) = 0.7211 Prob>chi2 = 0.868

28 . oneway average_heart_rate bb_status

Source	Analysis of variance			F	Prob > F
	SS	df	MS		
Between groups	2166.96794	3	722.322648	11.26	0.0000
Within groups	13987.0517	218	64.1607878		
Total	16154.0197	221	73.0951117		

Bartlett's equal-variances test: chi2(3) = 1.4962 Prob>chi2 = 0.683

29 .

30 . * post-hoc 事後検定

31 . pwmean sleep_hours, over(bb_status) mcompare(tukey)

Pairwise comparisons of means with equal variances

Over: **bb_status**

	Number of comparisons
bb_status	6

sleep_hours	Contrast	Std. err.	Tukey [95% conf. interval]	
bb_status				
HIGH vs GOOD	-.3882413	.3334681	-1.251576	.4750933
LOW vs GOOD	-.0760957	.2886241	-.8233311	.6711397
VERY_LOW vs GOOD	-.9960684	.3822524	-1.985703	-.0064333
LOW vs HIGH	.3121456	.3416937	-.5724847	1.196776
VERY_LOW vs HIGH	-.607827	.4237542	-1.704908	.4892544
VERY_LOW vs LOW	-.9199726	.389449	-1.928239	.088294

```
32 . pwmean average_heart_rate, over(bb_status) mcompare(tukey)
```

Pairwise comparisons of means with equal variances

Over: **bb_status**

	Number of comparisons
bb_status	6

average_heart_rate	Contrast	Std. err.	Tukey [95% conf. interval]	
bb_status				
HIGH vs GOOD	.5367733	1.514636	-3.384553	4.4581
LOW vs GOOD	5.518423	1.310951	2.124427	8.912418
VERY_LOW vs GOOD	8.066508	1.736217	3.571516	12.5615
LOW vs HIGH	4.981649	1.551997	.9635967	8.999702
VERY_LOW vs HIGH	7.529734	1.924721	2.546714	12.51276
VERY_LOW vs LOW	2.548085	1.768905	-2.031533	7.127703

```
33 . oneway steps bb_status, bonferroni
```

Source	Analysis of variance			F	Prob > F
	SS	df	MS		
Between groups	2636633.64	3	878877.878	0.27	0.8446
Within groups	701074397	218	3215937.6		
Total	703711030	221	3184212.81		

Bartlett's equal-variances test: $\chi^2(3) = 0.7211$ Prob> $\chi^2 = 0.868$

Comparison of steps by bb_status (Bonferroni)				
Row Mean- Col Mean	GOOD	HIGH	LOW	
HIGH	-230.828 1.000			
LOW	-240.869 1.000	-10.0411 1.000		
VERY_LOW	-147.588 1.000	83.2401 1.000	93.2812 1.000	

```

34 .
35 . * Body Battery分數化
36 . gen bb_score=.
    (222 missing values generated)

37 . replace bb_score=4 if body_battery_status=="HIGH"
    (43 real changes made)

38 . replace bb_score=3 if body_battery_status=="GOOD"
    (80 real changes made)

39 . replace bb_score=2 if body_battery_status=="LOW"
    (70 real changes made)

40 . replace bb_score=1 if body_battery_status=="VERY_LOW"
    (29 real changes made)

41 .
42 . * Multiple Regression
43 . reg bb_score sleep_hours deep_sleep_ratio rem_ratio

```

Source	SS	df	MS	Number of obs	=	222
Model	4.29957084	3	1.43319028	F(3, 218)	=	1.63
Residual	192.155384	218	.881446716	Prob > F	=	0.1843
				R-squared	=	0.0219
				Adj R-squared	=	0.0084
Total	196.454955	221	.888936448	Root MSE	=	.93885

bb_score	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
sleep_hours	.0427521	.0436915	0.98	0.329	-.0433596	.1288639
deep_sleep_ratio	-.7872425	.8115177	-0.97	0.333	-2.386667	.8121824
rem_ratio	-1.562093	.9093377	-1.72	0.087	-3.354311	.2301261
_cons	2.801303	.4122758	6.79	0.000	1.988747	3.61386

```

44 . outreg2 using regression_final.doc, replace word ctitle(Model 1 Sleep)
    regression_final.doc
    dir : seeout

```

```

45 .
46 . reg bb_score steps average_heart_rate

```

Source	SS	df	MS	Number of obs	=	222
Model	21.1800313	2	10.5900156	F(2, 219)	=	13.23
Residual	175.274924	219	.800342117	Prob > F	=	0.0000
				R-squared	=	0.1078
				Adj R-squared	=	0.0997
Total	196.454955	221	.888936448	Root MSE	=	.89462

bb_score	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
steps	4.21e-06	.0000337	0.12	0.901	-.0000623	.0000707
average_heart_rate	-.0361943	.0070389	-5.14	0.000	-.0500669	-.0223217
_cons	5.236554	.5248539	9.98	0.000	4.202143	6.270965

47 . outreg2 using regression_final.doc, append word ctitle(Model 2 Activity)
regression_final.doc
 dir : seeout

48 .

49 . reg bb_score sleep_hours deep_sleep_ratio rem_ratio steps average_heart_rate

Source	SS	df	MS	Number of obs	=	222
Model	25.5493988	5	5.10987976	F(5, 216)	=	6.46
Residual	170.905556	216	.791229427	Prob > F	=	0.0000
				R-squared	=	0.1301
				Adj R-squared	=	0.1099
Total	196.454955	221	.888936448	Root MSE	=	.88951

bb_score	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
sleep_hours	.0233895	.0421622	0.55	0.580	-.0597126	.1064915
deep_sleep_ratio	-.3803572	.7736473	-0.49	0.623	-1.905222	1.144507
rem_ratio	-2.016588	.8690942	-2.32	0.021	-3.729579	-.3035967
steps	.000012	.0000343	0.35	0.727	-.0000555	.0000795
average_heart_rate	-.0374024	.0072596	-5.15	0.000	-.0517111	-.0230937
_cons	5.600423	.6904544	8.11	0.000	4.239532	6.961313

50 . outreg2 using regression_final.doc, append word ctitle(Model 3 Full)
regression_final.doc
 dir : seeout

51 .

52 . reg bb_score sleep_hours steps average_heart_rate, beta

Source	SS	df	MS	Number of obs	=	222
Model	21.1803064	3	7.06010214	F(3, 218)	=	8.78
Residual	175.274649	218	.804012149	Prob > F	=	0.0000
				R-squared	=	0.1078
				Adj R-squared	=	0.0955
Total	196.454955	221	.888936448	Root MSE	=	.89667

bb_score	Coefficient	Std. err.	t	P> t	Beta
sleep_hours	.0006528	.0352858	0.02	0.985	.0012339
steps	4.32e-06	.0000344	0.13	0.900	.0081841
average_heart_rate	-.036164	.0072424	-4.99	0.000	-.3279332
_cons	5.229797	.6404367	8.17	0.000	.

53 . outreg2 using regression_final.doc, append word ctitle(base Model Beta)
regression_final.doc
 dir : seeout

54 .

end of do-file

55 .